Solving Exponential and Logarithmic Equations

Solving exponential equations with logarithms, solving logarithmic equations by rewriting them in exponential form, and using the logarithm laws to rewrite an equation before solving it

Before you start.

- A logarithm is an exponent: $\log_b M = k$ and $b^k = M$ say the same thing. Every problem here is one of those two sentences rewritten as the other.
- When the unknown sits in the exponent and both sides can be written with the same base, match the exponents. When they cannot, take a logarithm of both sides and use $\log(M^k) = k \log M$.
- Change of base: $\log_b M = \dfrac{\log M}{\log b}$, which is what makes a calculator answer possible.
- A logarithm is only defined for a positive argument, so every candidate root of a logarithmic equation must be checked in the original equation.
- Give a decimal answer only where the problem asks for one, and state the unit it carries.

1. A culture of yeast triples in size every hour, and it is measured as a multiple of its starting size. The model is $3^x = M$, where x is the number of hours elapsed. After how many hours is the culture 81 times its starting size?

Answer: ________________ hours

2. A bacterial population doubles once per day, so after x days it is 2^x times its starting size. Solve $2^x = 5$ to find when the population reaches 5 times its starting size. Give the answer to the nearest thousandth of a day.

Answer: ________________ days

3. A doubling-time model records the elapsed time in hours as $t = \log_2 n$, where n is the current size as a multiple of the starting size. Rewrite the equation in exponential form and find n when $t = 5$.

Answer: ________________

4. Rewrite $\log_3(9x^4)$ as a sum of a number and a multiple of $\log_3 x$, for $x > 0$. Show which logarithm law you use at each step.

Answer: ________________

5. A radioactive sample has a mass of 200 grams and a half-life of 6 years, so its mass after t years is $200\left(\frac{1}{2}\right)^{t/6}$ grams. After how many years does 25 grams remain?

Answer: ________________ years

6. An account opens with 1500, in dollars, and grows by four percent each year, so the balance is multiplied by 1.04 annually and after t years it is $1500(1.04)^t$ dollars. After how many years does the balance reach 2000 dollars? Give the answer to the nearest hundredth of a year.

Answer: ________________ years

7. Solve $\log x + \log(x-3) = 1$, where log is the common logarithm, base 10. Condense the left side first, then rewrite in exponential form. State every candidate root and say which one the equation rejects, and why.

Answer: ________________

8. Condense $2\log_2 x - \log_2 3$ into a single logarithm with base 2, for $x > 0$. This is the step that turns an equation of that shape into one you can rewrite in exponential form.

Answer: ________________

9. To solve $4^{x+1} = 32$, Dev divided 32 by 4 and set $x+1$ equal to the quotient. Explain in one sentence why dividing the two numbers is not a legal step here, then solve the equation correctly by writing both sides as powers of 2.

Answer: ________________

10. Solve $2\log x = \log(x+6)$, where log is the common logarithm. Condense the left side, rewrite in exponential form, and then state which of the two candidate roots the equation rejects and why.

Answer: ________________